

Chapter 49 — Anomalies of the Kidney and Collecting System — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Incidence ladder:** horseshoe 1:400 > ectopia 1:900 > URA 1:1200 > crossed 1:2000 > BRA 1:5000.
- **Bilateral renal agenesis (BRA)** fatal from **pulmonary hypoplasia**; lung window 16–26 weeks.
- **Unilateral renal agenesis (URA):** genital anomaly **females 25–50% vs males 10–15%.**
- **Absent vas** → **ipsilateral renal agenesis in 79%.**
- **Solitary kidney:** annual blood pressure (BP) + microalbumin; >30 mg/24 h = injury.
- **Crossed ectopia 90% fused;** inferior ectopia commonest (two-thirds).
- **Horseshoe:** isthmus below inferior mesenteric artery (IMA); calyces point posteriorly; Turner 60%.
- **Horseshoe pelviureteric junction (PUJ) obstruction 13–34%;** Wilms 1.76–7.93× expected.
- **Renal artery aneurysm (RAA):** treat if >2 cm, expanding, pregnancy, rupture, symptoms.
- **Calyceal diverticulum:** computed tomography (CT) or magnetic resonance urography (MRU); treat only if symptomatic or stone.

ANOMALIES OF RENAL NUMBER

- **BRA:** 1 in 4800–5000; **75% male;** 40% stillborn.
 - **First-degree relatives:** renal anomaly risk **1000×** population → screen parents, siblings by ultrasound.
 - **Anuria >24 h, no bladder distension** → suspect BRA; all neonates void <24 h.
 - **Gold standard if ultrasound equivocal:** **99mTc-DMSA (dimercaptosuccinic acid)** — no uptake anywhere = BRA.
 - **Pathognomonic:** Potter facies + oligohydramnios; amnion nodosum **not** pathognomonic.
 - **Potter facies:** infraorbital fold, blunt nose; ears only appear low set.
 - **"Lying down" adrenal:** flattened orthotopic adrenal = absent kidney.
 - **Fetal urine >90%** of third-trimester amniotic fluid.
 - **Undescended testes 43%** in BRA; gastrointestinal anomalies 60%.
 - **RAFT trial (serial amnioinfusion <26 weeks):** 94% live births; 35% survived to discharge; stopped early.
 - **Amnioinfusion for BRA = experimental;** institutional review board oversight only.
 - **URA: 1 in 1200;** male 1.8:1; left; autosomal dominant, 50–90% penetrance.
 - **Maternal diabetes** → **3×** agenesis/dysplasia; **VACTERL** (vertebral, anal, cardiac, tracheo-oesophageal, renal, limb) → **URA 30%.**
 - **Ipsilateral ureter absent 60%;** adrenal agenesis <10%.
 - **Contralateral:** vesicoureteric reflux (VUR) 30%; PUJ obstruction 11%; vesicoureteric junction (VUJ) obstruction 7%.
 - **Commonest associated urological problem with URA:** **VUR.**
 - **Confirm URA:** **DMSA** — also finds an ectopic or crossed kidney.
 - **Involved multicystic dysplastic kidney (MCDK) mimics URA:** curvilinear calcification; colonic flexure normally sited.
 - **Screening VCUG (voiding cystourethrogram) in URA:** benefit unproven → shared decision.
 - **Reproductive anomalies overall 20–40%;** male wolffian duct derivatives absent almost 50%.
 - **43% of females with internal genital anomaly have URA.**
 - **Infertility up to one-third** with agenesis + unicornuate uterus.
 - **Zinner:** seminal vesicle cyst + ipsilateral agenesis.
 - **OHVIRA / Herlyn-Werner-Wunderlich:** obstructed hemivagina + ipsilateral renal anomaly.
 - **MRKH (Mayer-Rokitansky-Küster-Hauser):** 1 in 4500 females; **type II (atypical, commoner) → renal anomaly 40–60%.**
 - **MURCS:** müllerian aplasia, renal aplasia, cervicothoracic (C5–T1) somite dysplasia.
 - **Magee timing:** type I <4 weeks (unicornuate); II early week 4 (didelphys, obstructed); III >4 weeks (isolated).
 - **Renal ultrasound if:** ear anomaly **plus** other malformation, or ≥2 anomalies.
 - **VACTERL → image all systems including spine (mandatory).**
 - **Pelvic ultrasound after thelarche, before menarche** in girls with URA.
 - **Renal aplasia (1:1300)** possibly commonest cause of congenital solitary kidney; MCDK 1:4300.
 - **Compensatory hypertrophy from 20 weeks** in about 90%.
 - **CAKUT (congenital anomalies of kidney and urinary tract):** 1 in 600.
 - **Nephron number range 200,000–1.8 million** (eightfold).
 - **KIMONO study:** about one-third renal injury by about 10 years.
 - **Solitary kidney:** about 50% dialysis by 30 (one cohort); population ESRD (end-stage renal disease) risk 4×.
 - **Solitary kidney + CAKUT → BP, albumin twice yearly.**
 - **GFR (glomerular filtration rate) <60 → BP, albumin, creatinine 2–4×/year.**
 - **GFR >60, no medication → creatinine every 5 years.**
 - **First-line for progressive solitary-kidney injury:** **ACE** (angiotensin-converting enzyme) inhibitor; limit salt, excess protein.
 - **Supernumerary kidney:** own capsule and blood supply; **60% caudal;** 50% common-stem ureter.
 - **Supernumerary:** Weigert-Meyer rule broken in 10%; 25% found only at autopsy.
- ### ANOMALIES OF RENAL ASCENT
- **Ascent complete 6–9 weeks.**
 - **Ectopia 1 in 900 (autopsy);** bilateral 10%; solitary ectopic 1 in 22,000.
 - **Commonest site:** pelvic (below aortic bifurcation).
 - **Hydronephrosis in ectopia:** 50% obstruction (PUJ 70%, VUJ 30%), 25% VUR, 25% malrotation.
 - **VUR 30–50%;** with unilateral ectopia reflux usually into orthotopic kidney.
 - **Genital anomalies about 15%** (series 15–45%); females 20–66%.
 - **Ectopic kidney: normal global, reduced differential function on DMSA.**
 - **Girl, continuous incontinence, "solitary kidney" → ectopic kidney with ectopic ureter.**

- **Cephalad ectopia:** near T10; **omphalocele**.
- **Thoracic kidney:** 1 in 13,000; **foramen of Bochdalek**, not intrapleural; left, male 2:1.
- **Thoracic kidney imaging:** CT or MRU.
- **Thoracic kidney in only 0.25% of diaphragmatic hernias.**
- **Operate on thoracic kidney only for respiratory distress.**

ANOMALIES OF RENAL FORM AND FUSION

- **Crossed ectopia:** second commonest fusion anomaly; fused 1:1000 live births, 1:2000 autopsy.
- **Left-to-right crossover 3:1;** non-fused about 10%; bilateral crossed rarest.
- **Six fused forms:** inferior (commonest) > sigmoid (second) ... superior ectopia (least).
- **Anterior pelvis = early fusion; medial pelvis = late fusion.**
- **VUR 20% crossed; 71% bilateral crossed.** Ectopic orifice about 3%.
- **Solitary crossed ectopia → imperforate anus 20%;** kidney L1–L3, ureter crosses above S2.
- **Stone in crossed ureter → pain on anephric side.**
- **Before paediatric surgery:** MRU + MRA (magnetic resonance angiography), no radiation.
- **Horseshoe:** 0.25%; male >2:1; lower-pole fusion 95%; isthmus L3/L4.
- **Isthmus usually anterior to aorta and IVC (inferior vena cava).**
- **Horseshoe blood supply:** single artery per moiety is the exception → image before surgery.
- **Horseshoe associations:** Turner 60% (commonest syndrome); neural tube defect 3%; VUR 8–32%.
- **Horseshoe:** urinary tract infection 30%; stones 20–80%; ≥50% asymptomatic.
- **Horseshoe PUJ obstruction causes:** high insertion, course over isthmus, crossing vessels.
- **Horseshoe tumours:** renal cell carcinoma (RCC) not increased; renal pelvic tumours increased.
- **Horseshoe Wilms:** 37% initially inoperable → preoperative chemotherapy; favourable histology.
- **Horseshoe transplant:** about 80% five-year graft survival.
- **Horseshoe stones:** calcium; metabolic defects as in other stone formers.
- **Horseshoe moieties:** asymmetric DMSA function in about two-thirds.

ANOMALIES OF RENAL ROTATION

- **Normal:** 90° ventromedial, weeks 6–9. Malrotation 1:939; male 2:1; **Turner**.
- **Weyrauch:** ventral = **commonest**; dorsal (180°) = **rarest**.
- **Only vessel course reveals direction of rotation.**

ANOMALIES OF RENAL VASCULATURE

- **Single renal artery 71–85%;** five segments; apical artery most variable.
- **Multiple** = >1 from aorta; **aberrant** = non-aortic origin; **accessory** = same segment.
- **Crossing vessel at a normal PUJ about 19–25%.**
- **RAA:** 0.1%; saccular 93%; 90% <2 cm; 75% asymptomatic; wreath calcification 30%.
- **RAA rupture 0.3% overall; 18% if >3 cm.**
- **Fusiform RAA at bifurcation = congenital.**
- **RAA hypertension 10–55%, renin-mediated.**
- **Multiple renal arteries do not raise hypertension risk.**
- **Congenital arteriovenous fistula (AVF):** female 3:1; upper pole 45%; bruit 75%; haematuria >75%.

- **Congenital AVF:** hypertension 50%, high-output failure 50%.
- **AVF gold standard:** selective arteriography or DSA (digital subtraction angiography) — cirroid, prompt venous filling.
- **Acquired AVF (biopsy) may close spontaneously; congenital rarely → embolise or resect.**

ANOMALIES OF THE RENAL COLLECTING SYSTEM

- **Bifid pelvis 10% — normal variant.**
- **Calyceal diverticulum:** 4.5/1000; stones almost 40%; milk of calcium layers.
- **Operate on diverticulum if:** enlarging with pain or infection, abscess, urosepsis, symptomatic stone.
- **Diverticulum options:** ureteroscopic, percutaneous (fulgurate lining) or laparoscopic marsupialisation.
- **Megacalycosis:** male 6:1; 12–20 calyces; pelvis undilated; **normal renogram → no surgery.**
- **Megacalycosis:** bilateral in males; segmental unilateral in females.
- **Hydrocalycosis:** dilated major calyx connected to pelvis; infundibular obstruction.
- **Infundibulopelvic stenosis:** bilateral; monitor yearly creatinine and GFR.
- **Bilateral anomalies → ESRD in over a third;** ACE inhibitor, protein restriction.
- **Infundibulopelvic stenosis surgery only if obstruction impairs function.**

CLASSIC TRAPS

- **Trap:** Potter features are malformations — **Actually:** deformations from absent amniotic fluid.
- **Trap:** adrenal absent in agenesis — **Actually:** present, flattened ("lying down").
- **Trap:** isolated preauricular pit needs renal scan — **Actually:** only with another anomaly.
- **Trap:** URA genital anomalies commoner in males — **Actually:** commoner in females.
- **Trap:** aberrant vessels cause ectopia — **Actually:** they result from ectopia.
- **Trap:** solitary kidney bars sport — **Actually:** motor vehicle accidents injure more kidneys.
- **Trap:** crossing vessel proves PUJ obstruction — **Actually:** present at 19–25% of normal PUJs.
- **Trap:** horseshoe calyces point medially — **Actually:** posteriorly (failed rotation).
- **Trap:** megacalycosis needs pyeloplasty — **Actually:** non-obstructive, normal renogram.
- **Trap:** horseshoe raises RCC risk — **Actually:** raises Wilms and renal pelvic tumours only.